



# MAXIMILIANO RODRIGUEZ

Award Winning Robotics Engineer & Multi-Platform Developer

## Professional Summary

Robotics Engineer & Software Developer | World Ranked #19 (WRO International 2023) & Venezuelan National Robotics Champion. Specialized in embedded systems (Arduino, Raspberry Pi), C#/C++ architecture, and hardware-software integration. Founder of Paper Vortex Studios.

## Contact

 +58 4120466522

 maxrsilvagni@gmail.com

 Av Octavio Camejo, Venezuela

 @maximilianorodriguezsilvagni

## Skills

- Systems Integration
- Rapid Prototyping
- Competitive Strategy
- Technical Presentation
- Game Engines
- 3D Modeling & Design
- Programming Languages
- Project Management
- Discipline & Focus

## Language

- English (C1)
- French (A2)
- Spanish (Native Language)

## Education

- Colegio Ítalo Venezolano "Angelo De Marta" | Puerto La Cruz, Anzoátegui, Venezuela High School (Expected Graduation: June 2027)
  - Current Standing: Sophomore (10th Grade)
  - Academic Performance: Consistently achieve top-tier marks in Mathematics, Physics, and Chemistry.
  - Language Proficiency: Native Spanish; Advanced Professional English, Simple French.

## Supplemental Technical Education

- TechCamp ADAKADEMY (2022): Intensive certification in Python.
- Unity Engine & C# (Self-Taught): 8+ months of independent study in Game Architecture and Object-Oriented Programming.
- Kurios Academy: Ongoing advanced robotics training and competitive strategy (2022–Present).
- Tech Academy (2025 – Present): Advanced robotics training; specialized in Arduino systems, sensor integration, and competitive hardware prototyping for the WRO.

## Projects & Leadership.



### Paper Vortex Studios (August 2025 - Present)

- Founded an independent studio to develop "The Other Side," a boss-rush survival game in Unity (C#).
- Engineered core gameplay mechanics using randomized value logic for dynamic enemy behavior and survival systems.
- Managing the full development cycle, from 2D environment design to object-oriented programming and team management.



### Golden Codex (WRO Team) (October 2025 – Present)

- World Finalist (2026): Will represent the Venezuela National Delegation in San Juan, Puerto Rico.

Developed ORBIT, an AI-powered social robot, designing a custom acoustic trilateralization system across 3 directional microphones, programming a conversational AI to autonomously redirect dialogue toward cultural topics, integrating a Raspberry Pi 5, expressive facial display, and 4-motor mobile platform. Targeted global cultural education, proposing ORBIT as a scalable Business model for museums and public spaces seeking AI-driven cultural outreach.



### Tomorrow's Innovators (WRO Team) (March 2023 – December 2024)

- World Finalist (2023): Represented the Venezuela National Delegation in Panama City, Panama; achieved 19th Place Worldwide and 4th Place in the Americas.

Developed a real-time automated tide monitoring system, reducing maritime accident risk by processing multi-depth sensor data on an Arduino UNO to trigger color-coded LED navigation alerts (green/yellow/red), by designing submerged cable terminals, programming analog signal thresholds in C++, and building a water-pump-driven simulation prototype. Targeted global coastal navigation safety and coral reef preservation, proposing a scalable deployment model for bays and waterways with significant tidal variation.

- National Finalist (2024): Secured a Top 4 ranking in the Venezuela National WRO Competition.

Engineered an autonomous seed-to-microgreen farming robot to achieving a fully automated 8-day growth cycle by integrating Arduino UNO, Raspberry Pi 4, stepper motor, servo, and LED grow lights programmed in C++. Built from recycled materials and aligned with UN SDG-12, delivering a scalable classroom tool to teach sustainable agriculture through robotics.

### Awards & Honors

#### Kurios Competition (2022–2025):

Achieved consecutive top-tier placements across four years of robotics competition, securing 1st Place Overall in 2022 and 2023, followed by 2nd Place Overall in 2024 and 2025.



### Martial Arts & Discipline

#### Angelo Escudero Taekwondo Academy | 2018 – Present

- Rank: Red Belt with One Black Stripe.
- Experience: 8 years of consistent training in high-performance martial arts.
- Key Skills: Developed advanced physical coordination, mental focus, and the discipline required to balance elite athletic training with high-level engineering projects.